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Footnotes:

- ^a STORM-PSR uses ± 3 frame tolerance (vs B3 ± 5).
- ^b Estimated from published figures or GitHub source code, not explicitly tabulated in the paper.
- ^c Kendall’s Tau baseline (0.91) from STEPs is measured on PennActions dataset, not IKEA ASM.
- ^d PTMA mcAP (cv) = cross-view; (cs) = cross-subject; (csv) = cross-subject-view. CSV is the most stringent protocol.

Table 1: IKEA ASM Dataset: Multi-task benchmark comparison against state-of-the-art methods. Results are presented as mean $\pm$ std across 3 seeds unless noted otherwise.

Task	Method	Metric	Score	Reference
Object Detection (Segmentation)				
AP@0.5	ResNeXt-101-FPN	AP@0.5	85.3%	[?]
AP@0.5	Mask R-CNN	AP@0.5	78.9%	[?]
AP COCO	ResNeXt-101-FPN	AP COCO	65.9%	[?]
AP COCO	Mask R-CNN	AP COCO	64.3%	[?]
POPW (ConvNeXt-Tiny)	—	AP@0.5	—	—
POPW (ConvNeXt-Tiny)	—	AP COCO	—	—
2D Body Pose Estimation (PCK)				
PCK@10px	MaskRCNN-ft (front view)	PCK@10px	64.3%	[?]
PCK@0.2	MaskRCNN-ft (front view)	PCK@0.2	88.0%	[?]
POPW (ConvNeXt-Tiny)	—	PCK@10px	—	—
POPW (ConvNeXt-Tiny)	—	PCK@0.2	—	—
Activity Recognition (Top-1 Accuracy)				
Top-1 (front view)	I3D RGB	Top-1	60.75%	[?]
Top-1 (all views)	I3D RGB	Top-1	63.09%	[?]
Top-1 (all views)	I3D RGB+Pose	Top-1	64.15%	[?]
Top-1 (overall)	P3D	Top-1	60.40%	[?]
Top-1 (top view, RGB)	I3D	Top-1	57.57%	[?]
Top-1 (all views, most relevant objects)	PC3D	Top-1	80.2%	[?]
Top-1 (all views, all objects)	2D-CNN	Top-1	79.7%	[?]
POPW (ConvNeXt-Tiny)	—	Top-1 (front)	—	—
POPW (ConvNeXt-Tiny)	—	Top-1 (all views)	—	—
Activity Recognition (mcAP)				
mcAP (csv)	PTMA	mcAP (csv)	84.47%	[?]
mcAP (cs)	PTMA	mcAP (cs)	86.99%	[?]
mcAP (csv)	MiniROAD	mcAP (csv)	79.94%	[?]
mcAP (cs)	MiniROAD	mcAP (cs)	80.84%	[?]
POPW (ConvNeXt-Tiny)	—	mcAP (csv)	—	—
POPW (ConvNeXt-Tiny)	—	mcAP (cs)	—	—
Temporal Localization (mAP@0.5)				
mAP@0.5	I3D (combined)	mAP@0.5	20.00%	[?]
mAP@0.5	ActionFormer (RGB-only)	mAP@0.5	21.49%	[?]
mAP@0.5	Gated SRM	mAP@0.5	21.77%	[?]
POPW (ConvNeXt-Tiny)	—	mAP@0.5	—	—
Phase Classification (Acc@1.0)				
Acc@1.0	STEPs	Acc@1.0	37.02%	[?]
POPW (ConvNeXt-Tiny)	—	Acc@1.0	—	—
Temporal Ordering (Kendall’s Tau)				
Kendall’s Tau	STEPs	Kendall’s Tau	0.91^c	[?]
POPW (ConvNeXt-Tiny)	—	Kendall’s Tau	—	—

Table 2: IndustReal Dataset: Multi-task benchmark comparison. Results are presented as mean $\pm$ std across 3 seeds unless noted otherwise.

Task	Method	Metric	Score	Reference
Assembly State Detection (ASD) — mAP@0.5				
mAP@0.5	YOLOv8m (COCO+synth+real)	mAP@0.5	83.8%	[?]
POPW (ConvNeXt-Tiny)	—	mAP@0.5	—	—
Activity Recognition (Top-1 / Top-5 Accuracy)				
Top-1	MViTv2 (Kinetics pretrained)	Top-1	66.45%	[?]
Top-5	MViTv2 (Kinetics pretrained)	Top-5	88.43%	[?]
POPW (ConvNeXt-Tiny)	—	Top-1	—	—
POPW (ConvNeXt-Tiny)	—	Top-5	—	—
Procedure Step Recognition (PSR) — ± 5 frame tolerance				
F1	B3 (rule-based)	F1	0.883	[?]
Precision	B3 (rule-based)	Precision	0.885	[?]
Recall	B3 (rule-based)	Recall	0.880	[?]
Procedure Step Recognition (PSR) — ± 3 frame tolerance				
F1	STORM-PSR	F1	0.901^a	[?]
POS	STORM-PSR	POS	0.812^a	[?]
POPW (ConvNeXt-Tiny)	—	F1 (± 5)	—	—
POPW (ConvNeXt-Tiny)	—	F1 (± 3)	—	—
POPW (ConvNeXt-Tiny)	—	POS (± 3)	—	—
Assembly State Recognition (F1@1)				
F1@1 (best seed)	SupCon+ISIL (ResNet-34)	F1@1	0.496^b	[?]
F1@1 (5-seed avg)	SupCon+ISIL (ResNet-34)	F1@1	0.848^b	[?]
mAP@R(+) (best seed)	SupCon+ISIL (ResNet-34)	mAP@R(+)	0.618^b	[?]
mAP@R(+) (5-seed avg)	SupCon+ISIL (ResNet-34)	mAP@R(+)	0.583^b	[?]
POPW (ConvNeXt-Tiny)	—	F1@1	—	—
Error Verification (IoU)				
IoU	StateDiffNet GCA (ResNet-34)	IoU	0.58^b	[?]
POPW (ConvNeXt-Tiny)	—	IoU	—	—
Head Pose Estimation (9-DoF MAE)				
9-DoF MAE	(First baseline — POPW establishes)	MAE	—	—

Table 3: Activity Recognition: Top-1 / Top-5 accuracy and mcAP comparison across methods.

Dataset	Method	Pretrained	Top-1	Top-5	Reference
IKEA ASM (33 classes)					
IKEA ASM	I3D RGB (front)	Kinetics	60.75%	—	[?]
IKEA ASM	I3D RGB (all views)	Kinetics	63.09%	—	[?]
IKEA ASM	I3D RGB+Pose (all views)	Kinetics	64.15%	—	[?]
IKEA ASM	P3D (overall)	Kinetics	60.40%	—	[?]
IKEA ASM	I3D (top view)	Kinetics	57.57%	—	[?]
IKEA ASM	PC3D (all views, most relevant)	ImageNet	80.2%	—	[?]
IKEA ASM	2D-CNN (all views, all objects)	ImageNet	79.7%	—	[?]
IKEA ASM	PTMA	—	84.47% ^d	—	[?]
IKEA ASM	MiniROAD	—	79.94% ^d	—	[?]
IKEA ASM	POPW (ConvNeXt-Tiny)	ImageNet	—	—	—
IndustReal (74 classes)					
IndustReal	MViTv2	Kinetics	66.45%	88.43%	[?]
IndustReal	POPW (ConvNeXt-Tiny)	ImageNet	—	—	—

Table 4: Object / Assembly State Detection: AP and AP@0.5 comparison.

Dataset	Method	Backbone	AP@0.5	AP COCO	Reference
IKEA ASM (7 object classes)					
IKEA ASM	ResNeXt-101-FPN	ResNeXt-101	85.3%	65.9%	[?]
IKEA ASM	Mask R-CNN	ResNet-50	78.9%	64.3%	[?]
IKEA ASM	POPW (ConvNeXt-Tiny)	ConvNeXt-Tiny	—	—	—
IndustReal (24 ASD states)					
IndustReal	YOLOv8m (COCO+synth+real)	YOLOv8m	83.8%	—	[?]
IndustReal	POPW (ConvNeXt-Tiny)	ConvNeXt-Tiny	—	—	—

Table 5: Pose Estimation: PCK metrics for IKEA ASM and 9-DoF MAE for IndustReal.

Dataset	Method	View	PCK@10px	PCK@0.2	Reference
IKEA ASM — 2D Body Pose (17 COCO keypoints)					
IKEA ASM	MaskRCNN-ft	front	64.3%	88.0%	[?]
IKEA ASM	MaskRCNN-ft	top	61.8%	85.2%	[?]
IKEA ASM	MaskRCNN-ft	side	59.4%	83.7%	[?]
IKEA ASM	POPW (ConvNeXt-Tiny)	front	—	—	—
IKEA ASM	POPW (ConvNeXt-Tiny)	all views	—	—	—
IndustReal — Head Pose (9-DoF from 17 keypoints)					
IndustReal	(First baseline)	egocentric	9-DoF MAE:	—	—

Table 6: Procedure Step Recognition: F1, Precision, Recall, and POS at ± 3 and ± 5 frame tolerances.

		Tolerance	F1	Precision	Recall	POS	Reference
B3 (rule-based)	IndustReal-PSR	± 5	0.883	0.885	0.880	—	[?]
STORM-PSR	IndustReal-PSR	± 3	0.901^a	—	—	0.812^a	[?]
POPW (ConvNeXt-Tiny)	—	± 5	—	—	—	—	—
POPW (ConvNeXt-Tiny)	—	± 3	—	—	—	—	—

Table 7: Efficiency comparison: model parameters, computational cost (GFLOPs), throughput (FPS), and hardware used for evaluation.

Dataset	Method	Params (M)	GFLOPs	FPS	Hardware
IKEA ASM Models					
IKEA ASM	PTMA	12.9	1.96	291	—
IKEA ASM	MiniROAD	10.5	1.08	325	—
IKEA ASM	ResNeXt-101-FPN	—	—	—	—
IKEA ASM	Mask R-CNN	—	—	—	—
IKEA ASM	I3D RGB	—	—	—	—
IKEA ASM	POPW (ConvNeXt-Tiny)	—	—	—	RTX 3060
IndustReal Models					
IndustReal	YOLOv8m (ASD)	—	—	—	—
IndustReal	MViTv2 (Activity)	—	—	—	—
IndustReal	B3 (PSR rule-based)	—	—	—	—
IndustReal	STORM-PSR	—	—	—	—
IndustReal	SupCon+ISIL (ResNet-34)	—	—	—	—
IndustReal	ActionFormer (Gated SRM)	27.70	83.28	21	GTX 1080
IndustReal	Gated SRM	33.55	121.09	16	GTX 1080
IndustReal	POPW (ConvNeXt-Tiny)	—	—	—	RTX 3060

Table 8: Target thresholds: the strongest baseline per metric that POPW must surpass to demonstrate improvement.

Dataset	Metric	Strongest Baseline	Score	Threshold to Beat
IKEA ASM	Object Detection AP@0.5	ResNeXt-101-FPN	85.3%	> 85.3%
IKEA ASM	Activity Top-1 (all views)	PC3D	80.2%	> 80.2%
IKEA ASM	Activity mcAP (csv)	PTMA	84.47%	> 84.47%
IKEA ASM	Activity mcAP (cs)	PTMA	86.99%	> 86.99%
IKEA ASM	Temporal Localization mAP@0.5	Gated SRM	21.77%	> 21.77%
IKEA ASM	Phase Classification Acc@1.0	STEPs	37.02%	> 37.02%
IKEA ASM	Temporal Order Kendall's Tau	STEPs (PennActions) ^c	0.91	> 0.91
IndustReal	ASD Detection mAP@0.5	YOLOv8m	83.8%	> 83.8%
IndustReal	Activity Top-1	MViTv2	66.45%	> 66.45%
IndustReal	Activity Top-5	MViTv2	88.43%	> 88.43%
IndustReal	PSR F1 (± 5)	B3	0.883	> 0.883
IndustReal	PSR F1 (± 3)	STORM-PSR	0.901	> 0.901
IndustReal	Assembly State F1@1	SupCon+ISIL (5-seed)	0.848	> 0.848
IndustReal	Error Verification IoU	StateDiffNet GCA	0.58	> 0.58